

When Attention Becomes Memory: KV-Caches, Fast Weights, and Synapses

DataForge 2026 — Pathway Track: Explain the Frontier | One-Page Concept Summary

1. The Problem: Memory That Grows With Every Token

Standard Transformer inference relies on a Key-Value (KV) cache to avoid redundant computation during autoregressive generation. At each decoding step, the model stores the key and value projections of every past token, producing a memory footprint that scales as $O(T \cdot d)$, where T is sequence length and d is model dimension (Vaswani et al., 2017). At context windows exceeding 100K tokens, this linear growth in T becomes a dominant infrastructure bottleneck in both memory capacity and memory bandwidth.

2. The Core Mechanism: Re-Association

Matrix multiplication is associative. In unnormalized (linear) attention, the per-query retrieval $\sum_{\tau} (q_{\tau} k_{\tau}^T) v_{\tau}$ can be algebraically regrouped as $q_{\tau} (\sum_{\tau} k_{\tau}^T v_{\tau}) = q_{\tau} S_{\tau}$ (Katharopoulos et al., 2020). The state matrix S_{τ} , a real-valued $d \times d$ matrix, accumulates all historical key-value associations into a single structure whose dimensions depend on model width, not sequence length — eliminating the linear dependency on T and yielding $O(d^2)$ memory that is constant regardless of how many tokens have been processed. This regrouping is exact for unnormalized linear attention only; it does not apply to standard softmax attention, where nonlinear normalization couples all keys and queries in a way that prevents factorization into a fixed-size state (Katharopoulos et al., 2020).

3. The Synaptic Correspondence

The incremental state update $S_{t+1} = S_t + k_t^T v_t$ takes the mathematical form of a Hebbian outer-product learning rule, $\Delta S_{ij} = k_{t,i} \cdot v_{t,j}$, directly paralleling ΔW_{ij} being proportional to $\text{pre}_i \times \text{post}_j$ (Hebb, 1949). This formal correspondence connects attention-derived memory to fast-weight systems studied since Schmidhuber (1992) and formalized by Schlag et al. (2021). This is stated as a mathematical structural equivalence, not a claim that biological synapses perform float32 outer products.

4. The Trade-Off: Fixed Memory vs. Bounded Capacity

Replacing a growing KV-cache with a fixed-size state matrix introduces a capacity constraint. When the number of stored associations exceeds the effective capacity of S , retrieval produces corrupted blends of stored values — associative crosstalk. For sparse, non-negative key vectors with components drawn independently from a Bernoulli distribution at activation probability p , expected pairwise key overlap is $E[k_i \cdot k_j] = D \cdot p^2$, where $E[k_i \cdot k_j]$ denotes the expected dot product between two stored keys and D is vector dimensionality. Reducing density p quadratically suppresses interference, but excessively sparse representations lose the capacity to encode distinct patterns, producing a U-shaped trade-off between interference and expressiveness.

Central claim: Fixed-state associative memory can replace sequence-length-growing token storage, but its finite capacity creates an interference trade-off — sparse, non-negative representations reduce expected overlap between stored associations ($E[\text{overlap}]$ scaling as p^2), while excessive compression or dense overlap causes retrieval failures.

5. Comparative Architecture Summary

Dimension	Softmax Attention + KV Cache	Linear Attention / Fast Weights	BDH (Sparse Hebbian State)
Memory scaling	$O(T \cdot d)$ — grows with T	$O(d^2)$ — constant w.r.t. T	$O(d^2)$ — constant w.r.t. T
Recall fidelity	Exact token-level retrieval	Approximate; interference under load	Approximate; mitigated by $\sim 5\%$ sparsity
Interference risk	None (explicit storage)	High with dense representations	Reduced by sparse non-negative activations
Interpretability	Token-level attention weights	Opaque dense state	Sparse activations enable per-neuron inspection

Sources: Vaswani et al. (2017); Katharopoulos et al. (2020); Schlag et al. (2021); Kosowski et al. (2025).

6. BDH: The Dragon Hatchling

Pathway's Dragon Hatchling architecture (Kosowski et al., 2025, arXiv:2509.26507) implements sparse Hebbian-like outer-product state updates within a GPU-friendly formulation (BDH-GPU). BDH employs non-negative sparse activations with observed average activity of approximately 5%, varying with input predictability rather than a fixed budget. The architecture incorporates a scale-free neuron graph and RoPE-modulated linear attention. At scales from 10M to 1B parameters, BDH matches GPT-2-class Transformer performance on language modeling and translation benchmarks while maintaining interpretable, sparse internal representations (Kosowski et al., 2025).

7. BDH-CQ: Recurrent Latent Reasoning

BDH-CQ (Pathway, 2026, arXiv:2608.09888) extends BDH's recurrent memory to enable latent reasoning: demonstrations update the recurrent state, and queries are solved via internal computation without generating explicit chain-of-thought tokens. Pathway reports that a 150M-parameter BDH-CQ configuration achieves pass@2 scores of approximately 21% (Low effort), 27% (Medium), and 30% (High) on ARC-AGI-1, at a developer-computed inference cost of \$0.0007 per task. These are Pathway-reported figures; independent reproduction was not performed for this submission. Internal recurrence iteration counts and update dynamics remain proprietary and undisclosed.

8. Limitations

Three boundaries constrain the claims above. First, fixed-capacity associative memory inevitably suffers interference when storage load substantially exceeds state dimensionality; no sparsity level eliminates this under extreme load. Second, BDH-CQ's internal recurrence mechanism is proprietary — the published effort tiers are benchmark results on ARC-AGI-1, not deployment metrics, and independent reproduction remains unavailable. Third, the interactive demonstration accompanying this summary uses a simplified educational model ($D=32$, $N=128$) that isolates the Hebbian outer-product update principle; this toy model is not equivalent to the full BDH architecture, which operates at $N=8192$ with RoPE, graph dynamics, and gated sparse MLPs.

Works Cited

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